Chapter 9
Design of Experiments in Metrology



One of the topics that has received relatively little attention in the metrology literature is the statistical design of experiments (DOEx). A premise of statistical DOEx is that efficiency is paramount in terms of the resources (cost, time, personnel, equipment, etc.. . .) used to perform an experiment. In the case of metrology studies, individual runs in an experiment may be very costly, and the experiment must be designed in such a way that information regarding the measurement process is obtained efficiently. In this chapter we present a variety of experimental designs and approaches to efficiently developing, evaluating, and optimizing a measurement process.

9.1 Introduction

Metrology experiments will typically be used in the design and optimization of a new measurement process or in the evaluation of an existing measurement process. To design and optimize a new measurement process, factorial and fractional factorial designs (DOEx) will often be used to identify factors and quantify their effect on the measurement. To evaluate an existing measurement process, an analysis of variance (ANOVA) approach will often be used to estimate sources of variation that affect the measurement. In this chapter the basics of these types of designs are presented along with case studies to illustrate the approach. The strategies for experimental design presented here will rely on historical knowledge of the measurement process, careful determination of the objectives of the experiment, and sample size considerations to guide the choice of experimental design.

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9.2 Factorial Experiments in Metrology

Section 6.4 introduced the notion of an indirect measurement as a measurement in which the value of the measurand is functionally related to the values of other measured variables. The measurement equation was represented by (Eq. 6.29):

$$y = f(X_1, X_2, \ldots, X_N).$$

Following the approach for indirect measurements presented in Chap. 6, the measurement equation is approximated via the Taylor series expansion:

$$\begin{aligned} y = f(X_1, X_2, \ldots, X_N) &\cong \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_N X_N \\ &= \beta_0 + \sum_{i=1}^{N} \beta_i X_i. \end{aligned} \tag{9.1}$$

In this relationship, the input variables, the $X_i s$, are the sources of variation affecting the measurand. The coefficients, the $\beta_i' s$, correspond to the partial derivatives from the Taylor series expansion (see Eq. 6.34). In some cases, the form of f will not be well known and must be determined experimentally. The DOEx approach can be used to efficiently identify input variables and determine the relationship between the measurand and the identified variables. A vast literature exists on the topic of DOEx (see, for example, Box et al. 2005 and Montgomery 2017), but little attention has been given to the use of DOEx in metrology. In metrology, the $X_i' s$ may include physical quantities such as voltage, equipment factors, test setup factors, environmental factors, or human factors that may have an impact on the outcome of the measurement.

Factorial DOEx attempts to systematically identify important factors and interrelationships between factors and to quantify their effect on the measurand. The approach first attempts to identify the important $X_i' s$ and associated $\beta_i' s$ in Eq. (9.1). A second experiment may then be required to improve the estimates of the $\beta_i' s$ and to check for the presence of second order terms. This sequential approach to experimentation may often be more efficient than a single large experiment. With results from a well-planned DOEx, the measurement process can be designed or optimized by reducing the effect of any variables that increase measurement uncertainty. The analysis may also point to improvements in equipment, materials, or procedures that reduce measurement uncertainty.

The sections that follow below outline a DOEx planning strategy for factorial experiments that we use in our work. The four basic steps are: (1) Select the appropriate measurand(s) and determine the objective of the experiment, (2) Select factors to be included in the experiment, (3) Select factor levels, and (4) Choose the design pattern. If the purpose of the experiment is to characterize an established measurement process, the methods of ANOVA that follow in Sect. 9.3 should be considered in addition to the factorial design strategy described in this section.

9.2.1 Defining the Measurand and Objective of the Experiment

The first step in any metrology experiment is to carefully define the measurand of interest. The definition of the measurand should include, in detail, how the measurement will be made, and any special physical conditions present during the taking of the measurement. The GUM specifies that "the measurand should be defined in sufficient detail that any uncertainty arising from its incomplete definition is negligible in comparison with the required accuracy of the measurement." This first step in the DOEx process should also include a statement of the objective of the study. The objective could be to (1) identify a measurement equation including inputs and coefficients, (2) optimize a measurement process given either a known or unknown measurement equation, or (3) perform an uncertainty analysis based on a known measurement equation.

9.2.2 Selecting Factors to Incorporate in the Experiment

Given a measurand and the objective of the experiment, the second step is to list all factors that might impact the measurement. Expert knowledge, previous experimental results, historical data analysis, literature research, brainstorming, cause and effect diagramming, etc. may be used to develop a list of potential factors. As an input to the measurement equation, each factor may itself be thought of as a measurand.

The next step is to determine which factors can be controlled experimentally. For each of the factors deemed controllable, it must be determined if the factor can be measurable with sufficient accuracy. The possibility of interactions between factors and how that might impact the measurand should also be considered.

Finally, it must be determined how the factors will be handled during the experiment. The factors identified in the steps above could be *controlled* at various levels, *held constant*, *left uncontrolled but monitored*, or *ignored* during the experiment. Logistics, costs, and suspected impact will determine which of the potential factors will be carried forward into the metrology experiment.

9.2.3 Selecting Factor Levels and Design Pattern

Next, the factor levels and design pattern for the experiment should be chosen. Factorial design patterns under consideration should include full factorials, in which the experimental trials consist of all possible combinations of factor settings, and fractional factorials, in which the experimental trials consist of a carefully chosen subset of the corresponding full factorial design.

Table 9.1 Choosing type of design

If the objective is	Use this type of design
Identify the important factors affecting a measurand. This information may be used to design a measurement process and identify the form of the resulting measurement equation	Depending on the number of factors to be investigated, a full factorial or fractional factorial, usually with factors at two levels
Estimate coefficients of a measurement equation and optimize the measurement process	Usually a full factorial with factors at two or three levels. Fewer factors than in the first stage
For an established measurement process, estimate components of variance and quantify measurement uncertainties	Factorial design or an ANOVA-based design with a combination of random and fixed factors in a crossed or nested classification

The choice of factor levels and design pattern will depend on the objective of the experiment. Table 9.1 lists the most common objectives and the experimental plans that correspond to those objectives. After the type of design is determined, the low and high levels for each factor should be selected. Any theoretical relationships between the experimental factors and the measurand should be used to select physically meaningful levels. Levels for each factor could also be determined from previous experience with the same factors. The selected levels should be bold, yet meaningful, within the context of the measurement process being studied. For each qualitative factor, the levels are simply the different categories within that factor. For example, if the qualitative factor is "Operator," the operators used in the experiment define the levels.

Based on the objective, the recommended choice of experimental design will be a full factorial, a fractional factorial, or an ANOVA-based design with random and possibly fixed effects.

For a thorough discussion of the types of designs, models, and analyses associated with full factorial designs and mixed-level factorials, see Montgomery (2017). For two-level fractional factorials and second order designs, see Box et al. (2005), and for random and mixed effects models and estimation of variance components, see Searle and Gruber (2016).

The full factorial and fractional factorial design and analysis approach is illustrated with case studies that follow in Sects. 9.2.4 and 9.2.5. The random and fixed effects design and analysis approach is illustrated with case studies that follow in Sects. 9.3.4 and 9.3.5.

9.2.4 Analysis of CMM Errors via Design of Experiments (2^4 Full Factorial)

The first case study demonstrates the design and analysis of a full factorial design. The experiment investigated the uncertainty of a coordinate measuring machine (CMM) contact probe measurement (Zhang, et al., 2002). The objective of the experiment was to quantify the uncertainty of the measurement with respect to several factors and to optimize use of the CMM.

Conclusion

Design of Experiments (DOEx) was used to analyze the uncertainty in a CMM's contact probe measurement. In the absence of a pre-defined measurement equation, a four-factor experiment was designed and performed to assess the effect of each factor on both the mean and standard deviation of the cylinder diameter measurement. The four factors included in the experiment were Sampling Speed, Part Size, Surface Form, and Probe Size. The objective of the experiment was to optimize use of the CMM in terms of minimizing uncertainty in the cylinder diameter measurements.

Analysis of the mean data showed that only Part Size and Surface Form impacted the mean cylinder diameter measurement. Both Sampling Speed and Probe Size did not affect the mean response, meaning that these factors did not introduce bias. Analysis of the standard deviations showed that only Surface Form and Probe Size impacted the standard deviation. Sampling Speed and Part Size did not affect the standard deviation, meaning these factors did not influence the uncertainty. The study indicated that the CMM performed better measuring smooth surface parts. When rough surface parts are measured, a larger probe size filters out the effects of the roughness, resulting in a smaller standard deviation. Depending on the intended use of the part, this may not be desired because it could mask the true condition of the part.

This experiment optimized use of the CMM in terms of these four factors, but it did not result in a formal uncertainty analysis. To perform a detailed uncertainty analysis, the CMM would be tested multiple times on multiple dimension using the optimal settings identified in this experiment. Both Type A and Type B uncertainties would be included in such a study. Methods for determining appropriate sample sizes for such an uncertainty study are detailed in Sect. 11.5.

9.2.5 Finite Element Method (FEM) Uncertainty Analysis via Design of Experiments (2^{7-3} Fractional Factorial)

The second case study demonstrates the design and analysis of a fractional factorial experiment. This experiment, taken from Fong et al. (2014), was part of an "uncertainty analysis of the finite element method-based solution of a proposed base design of a magnetic resonance imaging (MRI) RF coil." Figures in the Fong et al. (2014) reference show the usage and dimensions of the birdcage coil investigated in this experiment.

The authors presented a "metrological approach" to the verification and uncertainty quantification of the FEM-based simulations by treating each simulation as a virtual experimental run. The response variable YC (the "measurand") was a dimensionless quantity corresponding to the non-uniformity of the magnetic field profile produced by a run of the FEM code. The objective of the experiment was to identify the key factors affecting this non-uniformity measure. A second experiment,

$$\text{YC} = f(X_1, X_2, X_3) \cong \beta_0 + \sum_{i=1}^{3} \beta_i X_i + \sum_{i=1}^{3} \sum_{j \geq i}^{3} \beta_{ij} X_i X_j. \tag{9.5}$$

An analysis of this second order model would be used to optimize and finalize the RF Coil design. Once the design was finalized, a formal measurement uncertainty analysis could be performed using actual runs of the MRI imaging setup. Both Type A and Type B uncertainties would be included in such a study. Methods for determining appropriate sample sizes in an uncertainty study are detailed in Sect. 11.5.

The approach to fractional factorial designs presented here assumed all experimental factors were initially investigated at just two levels. In practice, fractional factorial designs may have factors with more than two levels, and factors with different numbers of levels. These more difficult cases are discussed by Montgomery (2017), and Addelman and Kempthorne (1961), who have catalogued orthogonal fractional factorial designs that allow estimation of all main effects with a minimum number of experimental trials.

9.2.6 Summary of Factorial DOEx Method

Full factorial and fractional factorial designs can be used in the design and optimization of a new measurement process or in the evaluation of an existing measurement process. These designs will often be used to identify the most important factors and to quantify their effect on the measurement. The results of these experiments can also be used to optimize a measurement process. The premise of statistical DOEx is that the resulting experimental design will be efficient in terms of the resources used to perform the study. Fractional factorial designs in particular provide an efficient framework for the investigation of many factors in a minimum number of experimental trials.

The design pattern and sample sizes for factorial and fractional factorial designs will depend on the number of factors, resolution of the design, and the desired precision in estimating repeatability. The degrees of freedom for repeatability will depend on the number of replicates of each experimental trial. The resolution must be chosen so that the important main effects and interactions can be estimated without serious confounding. Once the resolution of the design and the design pattern has been chosen, the number of replicates per trial can be chosen. It is recommended that the design have 20–30 degrees of freedom for repeatability, with 15 as minimally acceptable. More detailed recommendations regarding degrees of freedom and sample size are given in Sect. 11.5.

9.3 ANOVA Models in Metrology

Analysis of variance (ANOVA) is a statistical technique used to "identify and quantify individual *random* effects in a measurement so that they may be properly taken into account when the uncertainty of the result of the measurement is evaluated" (GUM, p. 98). It is used to estimate the variance components associated with random effects included in a measurement model. The objective is to gain an understanding of the sources of variance so that the measurement process may be improved. ANOVA is also used to estimate fixed effects that may contribute to the bias in a measurement. It differs from factorial design of experiments in that it is typically not used to identify factors or construct an indirect measurement equation.

ANOVA methods are often used in metrology when certifying reference standards by interlaboratory testing (GUM, p. 98). See ISO Guide 13528 (2015) for a detailed discussion of the statistical methods used in these comparisons. This testing typically consists of a reference standard being tested multiple times on multiple days at each of several laboratories. In this case, differences from day-to-day and differences between repeated measurements are treated as random. Differences between laboratories are treated as fixed. Another common use of ANOVA techniques is to assess the performance of a single measurement device via a Gauge Repeatability and Reproducibility (Gauge R&R) study, commonly used in manufacturing environments. These techniques can be used as key components of tester qualification and tester comparison studies.

ANOVA techniques include the use of many different statistical models (Searle and Gruber 2016). The sections that follow provide an introduction to some of the most commonly used ANOVA models in metrology, including simple random effects models and mixed effects models. In this context, the term "mixed effects model" means an ANOVA model containing both random and fixed effects. These models are particularly useful in measurement studies because they accommodate repeated measurements made on the same test unit under either fixed or varying conditions. The case studies presented below include analyses of both random effects and mixed effects model.

9.3.1 Random Effects Models

Factors with random effects are factors whose levels can be thought of as a random sample from some population (Kutner et al. 2005). As an example, consider a factory that manufactures steel plates. In this factory there are many operators responsible for measuring the thickness of the plates. Selecting a random sample of parts to be measured by a random sample of operators leads to a random effects model for the analysis. Each operator performs the thickness measurement on each plate n times. In this case both Part and Operator would be treated as having random

Table 9.9 Expected values of the mean squares for a two-way random effects model

Mean square	Expected value
$\mathrm{MS}_{\mathrm{Part}}$	$\sigma^2_{\mathrm{Error}} + n\sigma^2_{\mathrm{Part}\times\mathrm{Oper}} + on\sigma^2_{\mathrm{Part}}$
$\mathrm{MS}_{\mathrm{Oper}}$	$\sigma^2_{\mathrm{Error}} + n\sigma^2_{\mathrm{Part}\times\mathrm{Oper}} + pn\sigma^2_{\mathrm{Oper}}$
$\mathrm{MS}_{\mathrm{Part}\ \times\ \mathrm{Oper}}$	$\sigma^2_{\mathrm{Error}} + n\sigma^2_{\mathrm{Part}\times\mathrm{Oper}}$
$\mathrm{MS}_{\mathrm{Error}}$	$\sigma^2_{\mathrm{Error}}$

Table 9.10 Variance component estimates for a two-way random effects model

Variance component	Estimate
$\widehat{\sigma}^2_{\mathrm{Part}}$	$\frac{\mathrm{MS}_{\mathrm{Part}} - \mathrm{MS}_{\mathrm{Part}\times\mathrm{Oper}}}{on}$
$\widehat{\sigma}^2_{\mathrm{Oper}}$	$\frac{\mathrm{MS}_{\mathrm{Oper}} - \mathrm{MS}_{\mathrm{Part}\times\mathrm{Oper}}}{pn}$
$\widehat{\sigma}^2_{\mathrm{Part}\times\mathrm{Oper}}$	$\frac{\mathrm{MS}_{\mathrm{Part}\times\mathrm{Oper}} - \mathrm{MS}_{\mathrm{Error}}}{n}$
$\widehat{\sigma}^2_{\mathrm{Error}}$	$\mathrm{MS}_{\mathrm{Error}}$

represents the reproducibility variance. The part-to-part variance component σ^2_{Part} would not be of concern from the perspective of measurement uncertainty.

9.3.2 Mixed Effects Models

Mixed effects ANOVA models are used when both random and fixed effects are present in the measurement study. The difference between the two types of effects is sometimes subtle. For random effects, interest is in the component of variance associated with each effect. For fixed effects, interest is in the mean associated with each effect.

Consider a case where multiple calipers are being used to measure steel thickness and the goal is to quantify the difference between the calipers. Because the objective is to estimate differences between specific calipers, caliper would be considered a fixed factor. A two-factor mixed effects model would be used in the analysis, with one factor (Caliper) having fixed effects and one factor (Operator) having random effects. For simplicity, Part is excluded as a factor in this example, although it could easily be added as a second factor with random effects. The mixed effects model is thus defined as

$$y_{ijk} = \mathrm{Ave} + \mathrm{Operator}_i + \mathrm{Caliper}_j + (\mathrm{Operator} \times \mathrm{Caliper})_{ij} + \mathrm{Error}_{ijk} \qquad (9.7)$$

with $i = 1, \ldots, o$ operators, $j = 1, \ldots, c$ calipers, and $k = 1, \ldots, n$ measurements. Here, Ave is again the overall average measurement. The $\mathrm{Caliper}_j$ effects are fixed, while the remainder of the terms are modelled as normal random variables with mean zero and constant variances. For the mixed model expressed as in Eq. (9.7), the fixed factor Caliper assumes a different average for each of its levels, while the random factor Operator assumes the same variance for each of its levels.

Table 9.11 Expected values of the mean squares for a two-way mixed effects model

Mean square	Expected value
$\mathrm{MS}_{\mathrm{Caliper}}$	$\sigma^2_{\mathrm{Error}} + n\sigma^2_{\mathrm{Caliper}\times\mathrm{Oper}} + \mathrm{on}\frac{\sum \mathrm{Caliper}_j^2}{c-1}$
$\mathrm{MS}_{\mathrm{Oper}}$	$\sigma^2_{\mathrm{Error}} + cn\sigma^2_{\mathrm{Oper}}$
$\mathrm{MS}_{\mathrm{Caliper}\ \times\ \mathrm{Oper}}$	$\sigma^2_{\mathrm{Error}} + n\sigma^2_{\mathrm{Caliper}\times\mathrm{Oper}}$
$\mathrm{MS}_{\mathrm{Error}}$	$\sigma^2_{\mathrm{Error}}$

Table 9.12 Variance component estimates for a two-way mixed effects model

Variance component	Estimate
$\widehat{\sigma}^2_{\mathrm{Oper}}$	$\frac{\mathrm{MS}_{\mathrm{Oper}}-\mathrm{MS}_{\mathrm{Error}}}{\mathrm{cn}}$
$\widehat{\sigma}^2_{\mathrm{Caliper}\times\mathrm{Oper}}$	$\frac{\mathrm{MS}_{\mathrm{Caliper}\times\mathrm{Oper}}-\mathrm{MS}_{\mathrm{Error}}}{\mathrm{n}}$
$\widehat{\sigma}^2_{\mathrm{Error}}$	$\mathrm{MS}_{\mathrm{Error}}$

The expected mean squares and estimated variance components for the random factor and interaction are provided in Tables 9.11 and 9.12, respectively.

In the context of a measurement study, the primary focus is on the measurement-related variance terms $\sigma^2_{\mathrm{Error}}$, σ^2_{Oper}, and $\sigma^2_{\mathrm{Caliper}\times\mathrm{Oper}}$. The random effect terms would become part of the Type A combined standard uncertainty, providing estimates of both repeatability and reproducibility.

The estimate for each level of the fixed factor (Caliper) can be calculated through ordinary least squares. The resulting equation is (Kutner et al. 2005):

$$\widehat{\mathrm{Caliper}}_j = \bar{y}_{j.} - \bar{y}_{...} \tag{9.8}$$

with variance equal to

$$\widehat{\sigma}^2\left\{\widehat{\mathrm{Caliper}}_j\right\} = \frac{MS_{\mathrm{Caliper}\times\mathrm{Oper}}}{\mathrm{on}}. \tag{9.9}$$

The individual effect terms (9.8) and their corresponding variances (9.9) would be used to test for caliper differences that possibly introduce bias into the measurement.

9.3.3 *Underlying ANOVA Assumptions*

There are a few key assumptions to be aware of when performing an ANOVA:

1. Normality. Error terms are assumed to be normal, although slight deviations from normality are typically not critical.

2. Constant variance (homoscedasticity). This is a stronger assumption than normality. In a random effects or mixed effects model, it is assumed that the variance is the same at each level of a particular random effect.
3. Independence of the error terms. Plotting the residuals $e_{ijk} = \left(y_{ijk} - \bar{y}_{ij.}\right)$ is typically a good way to assess whether this assumption has been met. Lack of independence can dramatically affect an analysis and its interpretation.

Guidance on checking ANOVA assumptions is provided in Kutner et al. (2005). If these assumptions are not met, there may be additional modelling options. See Arendacka (2014) for dealing with non-constant variance (heteroscedasticity) and correlations (non-independence) between the random effects.

9.3.4 Gauge R&R Study (Random Effects Model)

Gauge R&R studies are frequently used in manufacturing to assess the amount of variation in measured data that can be attributed to the measurement process. A typical Gauge R&R study consists of multiple parts being measured multiple times by multiple operators, to allow estimation of both repeatability and reproducibility. More details on the structure and analysis of Gauge R&R studies are provided in Vardeman and Jobe (2016). A case study is presented in this section of a standard Gauge R&R study. It is followed by discussion and a proposal for designing Gauge R&R studies to improve the estimate of reproducibility.

A Gauge R&R study was performed to assess the uncertainty of a high-precision optical CMM used to inspect dimensions of a component assembled within the NSE. The study consisted of three operators ($o = 3$) measuring four manufactured parts ($p = 4$) three times each ($n = 3$) for a total of 36 measurements. The data appear in Table 9.13.

The graphical analysis in Fig. 9.13 shows that Parts 1 and 3 have very different lengths than Parts 2 and 4, but that the Operators are similar and repeated measurements of the same part by the same operator are similar.

The ANOVA associated with the study appears in Table 9.14. In this analysis, Part and Operator were both treated as random factors, for which the equations in Table 9.10 apply.

Equating mean squares to their expected values (Table 9.10) and solving for the variance components, the estimates are

$$\widehat{\sigma}^2_{\text{Error}} = 108.5$$

$$\widehat{\sigma}^2_{\text{Part}\times\text{Oper}} = \frac{136.8 - 108.5}{3} = 9.4$$

A second observation regards another aspect of the W-S approximation for this problem. Ballico (2000) showed that the W-S approximation is not a good approximation when an uncertainty term with many degrees of freedom is added to a dominant uncertainty term with a small number of degrees of freedom. This is often the case with Gauge R&R studies designed according to typical recommendations (see, for example, Down et al. 2010). The MS_{Oper} term in the calculation of υ_{Repr} will often be larger, with fewer degrees of freedom, than the MS_{Error} term, which typically has more degrees of freedom.

Reproducibility is defined as the uncertainty in a measurement system over time, under conditions that are subject to change. The standard Gauge R&R study typically looks at variation of a measurement system over a relatively short period of time, with only the operator being varied. Other factors affecting the reproducibility of the measurement are often not included in these studies. The degrees of freedom analysis above indicates that the standard Gauge R&R study produces little information with respect to reproducibility. The degrees of freedom associated with repeatability, however, is much more likely to be adequate.

Our recommendation is therefore to use the Gauge R&R study, as typically designed, to estimate only the repeatability portion of uncertainty. An estimate of reproducibility should be based on data collected over time under different conditions. This could be achieved in various ways. First, the control chart methodology described in Sect. 11.1 could be used to monitor uncertainty in the measurement over time. Repeated measurements of a check standard over time would provide a good estimate of reproducibility. A handful of manufactured units (say 3–5) could also be set aside and repeatedly measured by multiple operators over time. These two potential sources of data would provide a far better estimate of reproducibility than that obtained from a short-term Gauge R&R study such as described above. After acceptable estimates of R&R are obtained, a complete uncertainty analysis can be performed, and test uncertainty ratios (see Sect. 5.2.1.1) can be used to assess gauge performance relative to product specifications.

With few operators in a short-term Gauge R&R study, Operators could be treated as a fixed effect. A simple check for differences between operators could then be made, and any significant differences could be addressed by reviewing measurement procedures and training.

It should be noted that the same design structure used for a Gauge R&R study can be used for comparing laboratories, with the factor "Operator" replaced with the factor "Laboratory," as in Suzuki et al. (2018).

9.3.5 Voltage Standard Uncertainty Analysis (Mixed Effects Model)

An uncertainty study was performed by measuring trigger output voltages (see Sect. 6.3.1) with an external standard Tektronix TDS 3024 digitizer. The uncertainty study was performed over a three-day period with 10 measurements taken each day

Error components together accounted for 95% of the total variation adjusted for voltage differences.

Bias and uncertainty were significantly different at each voltage level. As a result, a detailed uncertainty analysis would be done at each level of input trigger voltage. The uncertainty analysis would follow the approach given in Sect. 6.3.1 for the standard tested at 350 V. A separate expanded uncertainty (see Eq. 6.27), including the bias term, would be reported for each level.

9.3.6 Summary of ANOVA Method

ANOVA models, including random effects and mixed effects models, are useful for assessing the amount of variation that can be attributed to random components of a measurement process. ANOVA techniques provide a simple way to estimate variance components of the associated models. The relative magnitudes of the estimated variance components indicate where emphasis should be placed to reduce measurement uncertainty.

Sample sizes for ANOVA-based measurement studies should be based on a degrees of freedom argument. Each estimated variance component has a degrees of freedom associated with it, often computed using the Welch–Satterthwaite approximation. The degrees of freedom should always be reported along with the estimated variance component to assess the adequacy of the estimate.

A recommended goal in an ANOVA is to have 20–30 degrees of freedom for the error variance term, with 15 as minimally acceptable. Other variance components in an ANOVA will typically have far fewer degrees of freedom. It is recommended that the number of degrees of freedom for these components be built up over time through ongoing monitoring of the measurement system. More detailed recommendations regarding degrees of freedom and sample size are given in Sect. 11.5.

9.4 Related Reading

Metrology's guiding bodies have to date provided relatively little guidance on the use of design of experiments (DOEx) and ANOVA in uncertainty analyses. The JCGM 100 (GUM) has a brief discussion of basic ANOVA models and uses of ANOVA in metrology, but with no discussion of DOEx other than to say "a well-designed experiment can greatly facilitate reliable evaluations of uncertainty and is an important part of the art of measurement." The JCGM 101 and JCGM 102 have no mention of DOEx or ANOVA. The JCGM 103 "Developing and Using Measurement Models" (in draft form at this writing) discusses the use of experimental data to develop measurement models, but without discussion of DOEx.

ISO 3534-3, "Statistics - Vocabulary and Symbols - Part 3: Design of Experiments" defines the terms used in the field of design of experiments, although

compare repeatability and reproducibility. Less attention is given to the design of the Gauge R&R study. The authors recommend the study include at least six operators, 1020 parts, and two or three replicates. The book also includes a substantial bibliography of papers that will be helpful to practitioners of Gauge R&R and other ANOVA studies in metrology.

9.5 Exercises

1. When might a full factorial design be used in a measurement study? When might a fractional factorial design be used?
2. What are the various ways experimental factors can be handled during an experiment?
3. What is "confounding" of factors in a two-level fractional factorial design? How would one determine the confounding pattern for such a design?
4. Design a $2^{(6-2)}$ fractional factorial design for six factors at two levels. What is the highest resolution that can be achieved? Suppose it is suspected that the AB and CE two-way interaction terms are both important. How would this affect the design strategy?
5. Table 9.17 presents measurement data from a full factorial experiment in which moisture Gauges were used to measure the dewpoint of sealed containers at sub-zero temperatures.
   The experimental factors are: 3 Moisture Gauges, 3 Dewpoint Levels, and 2 Operators, with 2 replicates of each measurement. The standard dewpoint value in the table can be considered the "actual" value. The total number of experimental observations is $3 \times 3 \times 2 \times 2 = 36$.

   Explore the data graphically, including a main effects plot. What does the graphical analysis suggest? Perform an ANOVA treating Moisture Gauge and Dewpoint Level as fixed factors and Operator as a random factor. Which factor has the greatest effect on the dewpoint measurement? Which factors contribute to the measurement uncertainty? What is the estimated repeatability standard deviation? What is the total Type A standard uncertainty associated with the measurement?

   Compute the differences Delta = (Measured Value − Standard Value). Graph the main effects associated with this calculated value. Is the measurement biased? Perform an ANOVA treating Moisture Gauge and Dewpoint Level as fixed effects and Operator as a random effect. Which factors contribute most to the measurement bias?
6. When might a variance components design (ANOVA) be used in a measurement study? What is a mixed ANOVA model? What is the difference between factors that are crossed and factors that are nested?
7. What is the Welch–Satterthwaite approximation used for in ANOVA studies?
8. Table 9.18 gives the data resulting from a Gauge R&R study in which the length of a component piece part was measured. The nominal length of the part was

Table 9.18 Gauge R&R study data

Observation	Day	Operator	Part number	Replicate	Length (in)
1	1	1	1	1	0.970558
2	1	1	1	2	0.970183
3	1	1	1	3	0.969758
4	1	1	2	1	0.977658
5	1	1	2	2	0.977967
6	1	1	2	3	0.977933
7	1	2	1	1	0.970275
8	1	2	1	2	0.969833
9	1	2	1	3	0.969817
10	1	2	2	1	0.978125
11	1	2	2	2	0.978125
12	1	2	2	3	0.977908
13	2	1	1	1	0.970625
14	2	1	1	2	0.969758
15	2	1	1	3	0.970258
16	2	1	2	1	0.977833
17	2	1	2	2	0.978525
18	2	1	2	3	0.977600
19	2	2	1	1	0.970425
20	2	2	1	2	0.969667
21	2	2	1	3	0.970058
22	2	2	2	1	0.977833
23	2	2	2	2	0.978075
24	2	2	2	3	0.978292

associated with this measurement? What are the degrees of freedom for repeatability? Reproducibility?

9. What is an adequate number of degrees of freedom for estimating the error variance component (repeatability) in an ANOVA experiment? How can you increase the number of degrees of freedom to improve the estimate of reproducibility in a Gauge R&R study?
10. When would you use a Resolution III fractional factorial design?

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